

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 5: Parallel and Intersecting Lines

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	5 — Parallel and Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every item chains three or more relations, so a single mis-applied rule — equating co-interior angles instead of supplementing them, treating a non-parallel crossing as parallel, or halting before the last linear-pair step — lands you on a pre-computed wrong option. Do not write on this paper — mark answers on your answer sheet.

1. Two lines AB and CD cross at O, with rays OA, OC, OB, OD in order around the point. $\angle AOC = (2x + 18)^\circ$ and the adjacent angle $\angle AOD = (3x - 8)^\circ$ form a linear pair. Find the measure of $\angle BOC$.

- (A) 94° (B) 86°
(C) 34° (D) 46°

2. Two lines intersect at O. One of the four angles is $(5x - 4)^\circ$ and the angle vertically opposite it is $(3x + 30)^\circ$. Find the measure of the angle that forms a linear pair with these.

- (A) 81° (B) 17°
(C) 99° (D) 162°

3. Lines AB and CD intersect at O with $\angle AOC$ and $\angle BOD$ vertically opposite. $\angle AOC = (4x + 9)^\circ$ with $x = 12$. Ray OE lies inside $\angle BOD$ so that $\angle BOE = 31^\circ$. Find $\angle DOE$.

- (A) 57° (B) 31°
(C) 26° (D) 88°

4. Two lines cross at O. Reading once around, the four angles are $(3x)^\circ$, $(x + 40)^\circ$, $(3x)^\circ$, $(x + 40)^\circ$. After finding x, what is the measure of the larger of the two distinct angles?

- (A) 105° (B) 75°
(C) 35° (D) 115°

5. At the crossing of two lines, an angle is $(6x - 10)^\circ$ and its adjacent (linear-pair) angle is $(4x + 30)^\circ$. After finding the two angles, by how much does the larger exceed the smaller?

- (A) 16° (B) 8°
(C) 94° (D) 180°

6. Two lines meet at O. $\angle 1 = (7x - 16)^\circ$ and the angle vertically opposite $\angle 1$ is $(5x + 22)^\circ$. Find the measure of each angle in the OTHER vertical pair at O.

- (A) 117° (B) 19°
(C) 126° (D) 63°

7. Three rays OA, OB, OC leave point O and divide the full turn into $\angle AOB = (2x + 5)^\circ$, $\angle BOC = (3x - 15)^\circ$, and $\angle COA = (4x + 10)^\circ$. Find the largest of the three angles.

- (A) 85° (B) 105°
(C) 170° (D) 150°

8. Three rays from O split the full turn into angles $(2y)^\circ$, $(3y)^\circ$, and $(y + 60)^\circ$. After finding y, what is the supplement of the smallest of the three angles?

- (A) 100° (B) 150°
(C) 30° (D) 80°

9. Four rays from O make four angles around the point, in order: $(a)^\circ$, $(a + 10)^\circ$, $(a + 20)^\circ$, and $(a + 30)^\circ$. Find the largest of the four angles.

- (A) 105° (B) 75°
(C) 90° (D) 95°

10. Five equal angles fit exactly around a point O, sharing the common vertex. A ray OP bisects one of these five angles. What angle does OP make with the nearer arm of that angle?

- (A) 72° (B) 36°
(C) 45° (D) 18°

11. Rays OA, OB, OC, OD around point O make $\angle AOB = (3x)^\circ$, $\angle BOC = 90^\circ$, $\angle COD = (2x)^\circ$, $\angle DOA = (x + 30)^\circ$. Find $\angle AOB$.

- (A) 40° (B) 80°
(C) 120° (D) 70°

12. Three rays from O divide the full turn into $\angle 1 = (3x - 10)^\circ$, $\angle 2 = (x + 40)^\circ$, and $\angle 3 = (2x + 30)^\circ$. Find the measure of the MIDDLE-sized angle.

- (A) 140° (B) 90°
(C) 130° (D) 50°

13. A transversal cuts parallel lines p and q . A corresponding angle at p is $(5x - 12)^\circ$ and the matching corresponding angle at q is $(3x + 8)^\circ$. After finding x , find the linear-pair neighbour of that corresponding angle.

- | | |
|----------------|-----------------|
| (A) 38° | (B) 142° |
| (C) 10° | (D) 122° |

14. A transversal cuts parallel lines m and n . A corresponding angle at m is $(4x + 15)^\circ$ and the matching one at n is $(6x - 25)^\circ$. After finding x , find the co-interior (same-side interior) partner of that angle at the OTHER line's crossing.

- | | |
|----------------|-----------------|
| (A) 95° | (B) 20° |
| (C) 85° | (D) 105° |

15. Parallel lines l and m are cut by a transversal at P (on l) and Q (on m). At P an angle of 118° is formed. Find the angle at Q that is vertically opposite the corresponding angle of that 118° angle.

- | | |
|-----------------|-----------------|
| (A) 62° | (B) 236° |
| (C) 118° | (D) 90° |

16. A transversal cuts parallel lines a and b . A corresponding angle at a is $(2x + 30)^\circ$ and at b is $(4x - 10)^\circ$. After finding x , find the angle that forms a linear pair with the corresponding angle at b .

- | | |
|----------------|-----------------|
| (A) 70° | (B) 110° |
| (C) 20° | (D) 140° |

17. Parallel lines l and m are cut by a transversal. An angle of 64° is formed at the upper line l , opening toward the transversal on the interior side. Find the interior angle at m that lies on the SAME side of the transversal as the 64° angle.

- | | |
|-----------------|-----------------|
| (A) 116° | (B) 64° |
| (C) 26° | (D) 128° |

18. Parallel lines m and n are cut by a transversal. Corresponding angles are $(x + 25)^\circ$ at m and $(2x - 15)^\circ$ at n . After finding x , find the angle vertically opposite the corresponding angle at m .

- | | |
|-----------------|----------------|
| (A) 115° | (B) 40° |
| (C) 50° | (D) 65° |

19. Parallel lines l and m are cut by a transversal. A pair of alternate interior angles measure $(3x - 5)^\circ$ and $(x + 41)^\circ$. After finding x , find the co-interior (same-side interior) partner of one of these angles at its own crossing.

- | | |
|-----------------|----------------|
| (A) 64° | (B) 23° |
| (C) 116° | (D) 46° |

20. Parallel lines l and m are cut by a transversal. A pair of alternate exterior angles measure $(5x - 30)^\circ$ and $(3x + 10)^\circ$. After finding x , find the linear-pair neighbour of either alternate exterior angle.

- (A) 70° (B) 110°
(C) 20° (D) 40°

21. Parallel lines l and m are cut by a transversal. $\angle A$ on l and $\angle B$ on m are alternate interior angles, and $\angle A = (2 \cdot \angle B - 24)^\circ$. Find $\angle A$.

- (A) 24° (B) 48°
(C) 12° (D) 156°

22. Parallel lines l and m are cut by a transversal. Alternate interior angles measure $(7x - 4)^\circ$ and $(5x + 14)^\circ$. After finding the angle, find its co-interior partner at the same crossing.

- (A) 121° (B) 59°
(C) 9° (D) 31°

23. Parallel lines are cut by a transversal. An exterior angle at the upper line is 142° . Find the alternate interior angle at the lower line.

- (A) 142° (B) 38°
(C) 52° (D) 48°

24. Parallel lines l and m are cut by a transversal. Alternate exterior angles measure $(3x + 15)^\circ$ and $(4x - 5)^\circ$. After finding the angle, find the interior angle at the same crossing that forms a linear pair with it.

- (A) 75° (B) 20°
(C) 95° (D) 105°

25. Parallel lines are cut by a transversal. Co-interior angles measure $(3x + 20)^\circ$ and $(2x + 10)^\circ$. After finding x , find the alternate interior angle of the LARGER co-interior angle.

- (A) 70° (B) 110°
(C) 30° (D) 100°

26. Parallel lines are cut by a transversal. The two co-interior angles are in the ratio $7 : 2$. Find the smaller co-interior angle.

- (A) 140° (B) 20°
(C) 90° (D) 40°

27. Parallel lines are cut by a transversal. One co-interior angle is four times its same-side partner. Find the larger co-interior angle.

- (A) 144° (B) 36°
(C) 45° (D) 135°

28. Parallel lines are cut by a transversal. The two co-interior angles differ by 36° . Find the larger co-interior angle.

- (A) 72° (B) 108°
(C) 36° (D) 90°

29. Parallel lines are cut by a transversal. Co-interior angles measure $(5x)^\circ$ and $(4x - 9)^\circ$. After finding x , find the corresponding angle of the $(5x)^\circ$ angle at the other line.

- (A) 105° (B) 75°
(C) 21° (D) 115°

30. Parallel lines are cut by a transversal. At one crossing, an alternate-interior angle and a co-interior angle share the vertex and form a linear pair, in the ratio 3 : 2. Find the alternate-interior angle.

- (A) 72° (B) 108°
(C) 54° (D) 120°

31. Parallel lines a and b are cut by a transversal. Corresponding angles are $\angle P = (8x - 12)^\circ$ at a and $\angle Q = (5x + 21)^\circ$ at b . After finding the angle, find the linear-pair neighbour of $\angle Q$.

- (A) 76° (B) 11°
(C) 94° (D) 104°

32. Two lines intersect at O so that $\angle AOC = (7x)^\circ$ and the adjacent $\angle COB = (2x)^\circ$ form a linear pair. Ray OD bisects $\angle AOC$. Find $\angle DOC$.

- (A) 140° (B) 70°
(C) 40° (D) 20°

33. Parallel lines are cut by a transversal. Alternate interior angles measure $(2x + 9)^\circ$ and $(3x - 6)^\circ$. After finding x , find the co-interior partner of one of these angles at its crossing.

- (A) 39° (B) 15°
(C) 151° (D) 141°

34. Parallel lines are cut by a transversal. Co-interior angles measure $(x + 30)^\circ$ and $(2x)^\circ$. After finding x , find the vertically opposite angle of the LARGER co-interior angle.

- (A) 80° (B) 50°
(C) 130° (D) 100°

35. Two lines intersect at O . $\angle AOC = (4x + 5)^\circ$ and its vertically opposite $\angle BOD = (6x - 25)^\circ$. After finding the angle, find the linear-pair neighbour $\angle AOD$.

- (A) 65° (B) 15°
(C) 115° (D) 130°

36. Parallel lines p and q are cut by a transversal. Corresponding angles measure $(6x - 8)^\circ$ at p and $(4x + 22)^\circ$ at q . After finding x , find the co-interior partner of that angle.

- (A) 98° (B) 82°
(C) 15° (D) 108°

37. A transversal cuts two lines. A pair of corresponding angles measure 88° at the first line and 92° at the second line. What is the correct conclusion?

- (A) The two lines are parallel (B) The two lines are perpendicular
(C) The two lines are not parallel (D) The two lines are coincident

38. A transversal cuts two lines. A pair of co-interior (same-side interior) angles measure 95° and 86° . Are the lines parallel?

- (A) Parallel; the co-interior angles sum to 180°
- (B) Parallel; the angles are nearly equal
- (C) Perpendicular; one angle is near 90°
- (D) Not parallel; the co-interior angles sum to 181° , which exceeds 180°

39. A transversal cuts two lines. A pair of alternate interior angles measure 70° at one line and 76° at the other. What can you conclude?

- (A) The lines are parallel
- (B) The lines are perpendicular
- (C) The lines are not parallel
- (D) The lines meet at 6°

40. A transversal cuts two lines. A pair of co-interior angles measure 105° and 75° . What is the correct conclusion?

- (A) The lines are not parallel
- (B) The lines are parallel
- (C) The lines are perpendicular
- (D) Nothing can be decided

41. A transversal cuts two lines. A pair of corresponding angles each measure 102° . What is the correct conclusion?

- (A) The lines are parallel
- (B) The lines meet at 102°
- (C) Nothing can be concluded
- (D) The lines are perpendicular

42. Parallel lines l and m are cut by transversal t . $\angle 1$ corresponds to $\angle 2$; $\angle 2$ forms a linear pair with $\angle 3$; $\angle 3$ is vertically opposite $\angle 4$. If $\angle 1 = 124^\circ$, find $\angle 4$.

- (A) 124°
- (B) 56°
- (C) 34°
- (D) 236°

43. Three parallel lines l , m , n (in that order) are cut by a single transversal. At line l an interior angle of 73° opens toward the transversal. Find the interior angle at line n that lies on the SAME side of the transversal as that 73° angle.

- (A) 73°
- (B) 107°
- (C) 17°
- (D) 146°

44. Segment AB is parallel to segment CD . A point E lies between the two parallels, joined to B on AB and to C on CD , forming a 'Z' bend at E . $\angle ABE = 40^\circ$ and $\angle DCE = 65^\circ$. Find $\angle BEC$.

- (A) 105°
- (B) 25°
- (C) 75°
- (D) 140°

45. Segment AB is parallel to segment CD , with point E joined to B and to C . A line drawn through E parallel to both shows $\angle ABE = 58^\circ$ and $\angle DCE = 23^\circ$ lie on the SAME side of that helper line. Find $\angle BEC$.

- (A) 81°
- (B) 116°
- (C) 46°
- (D) 35°

46. Parallel lines l and m are cut by transversal t at P (on l) and Q (on m). At P the angle on the upper-right is 118° . Find the angle at Q on the lower-left that is co-interior with the 118° angle's corresponding angle.

- (A) 118° (B) 236°
(C) 62° (D) 28°

47. Segment AB is parallel to segment CD with point E between them, joined to B and C . $\angle ABE = 35^\circ$ and the bend angle $\angle BEC = 90^\circ$. Using a helper line through E parallel to both, find $\angle DCE$.

- (A) 125° (B) 45°
(C) 35° (D) 55°

48. Parallel lines l and m are cut by a transversal. The bisectors of the two co-interior angles are drawn. What is the angle between the two bisectors?

- (A) 45° (B) 100°
(C) 90° (D) 80°

49. Parallel lines a and b are cut by a transversal. A corresponding angle measures 84° . The bisector of the corresponding angle at line b is drawn. Find the acute angle this bisector makes with line b .

- (A) 84° (B) 48°
(C) 96° (D) 42°

50. Parallel lines l and m are cut by a transversal. An interior angle at l is 73° ; its co-interior partner $\angle Q$ at m is bisected. Find the acute angle the bisector of $\angle Q$ makes with line m .

- (A) 36.5° (B) 73°
(C) 53.5° (D) 107°

51. Two lines intersect at O with $\angle AOC = (7x)^\circ$ and adjacent $\angle COB = (2x)^\circ$ forming a linear pair. Ray OD bisects $\angle AOC$. Find $\angle DOB$.

- (A) 70° (B) 40°
(C) 140° (D) 110°

52. Parallel lines l and m are cut by a transversal. A pair of alternate interior angles each measure 96° , and each is bisected. What is the relationship between the two bisectors?

- (A) They are perpendicular (B) They meet at 96°
(C) They meet at 48° (D) They are parallel

53. Which ordered list could be the four angles, read once around the point, formed by two intersecting lines that are NOT perpendicular?

- (A) $65^\circ, 115^\circ, 115^\circ, 65^\circ$ (B) $65^\circ, 65^\circ, 115^\circ, 115^\circ$
(C) $90^\circ, 90^\circ, 90^\circ, 90^\circ$ (D) $65^\circ, 115^\circ, 65^\circ, 115^\circ$

54. Which statement is FALSE for two lines cut by a transversal?

- (A) Co-interior angles are equal whenever the two lines are parallel.
- (B) Alternate interior angles are equal whenever the two lines are parallel.
- (C) Vertically opposite angles are equal whether or not the lines are parallel.
- (D) Corresponding angles are equal whenever the two lines are parallel.

55. Which single condition does NOT by itself guarantee that two lines cut by a transversal are parallel?

- (A) A pair of co-interior angles are equal.
- (B) A pair of alternate interior angles are equal.
- (C) A pair of corresponding angles are equal.
- (D) A pair of co-interior angles sum to 180° .

56. A transversal cuts two parallel lines, and one of the eight angles is a right angle. How many of the eight angles are right angles?

- (A) 4°
- (B) 2°
- (C) 8°
- (D) 1°

57. A transversal crosses two parallel lines at exactly 90° . How many DISTINCT angle measures appear among the eight angles formed?

- (A) 1°
- (B) 2°
- (C) 4°
- (D) 8°

58. Line a is perpendicular to line b, and line c is parallel to line b. A transversal makes a 35° angle with line a. Find the acute angle the transversal makes with line c.

- (A) 35°
- (B) 55°
- (C) 90°
- (D) 125°

59. A transversal crosses line a at 90° . Line b is parallel to line a, and one angle the transversal makes with b is $(5x - 5)^\circ$. Find x.

- (A) 19°
- (B) 18°
- (C) 17°
- (D) 95°

60. Line a is perpendicular to line b, and line b is perpendicular to line c (all in one plane). A transversal makes 28° with line a. Find the acute angle the same transversal makes with line c.

- (A) 62°
- (B) 28°
- (C) 90°
- (D) 0°

Solutions — Mathematics (Class 7), Chapter 5: Parallel and Intersecting Lines — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	A	31	D	41	A	51	D
2	C	12	C	22	A	32	B	42	B	52	D
3	C	13	B	23	B	33	D	43	B	53	D
4	A	14	C	24	D	34	D	44	A	54	A
5	B	15	C	25	B	35	C	45	D	55	A
6	D	16	B	26	D	36	A	46	C	56	C
7	C	17	A	27	A	37	C	47	D	57	A
8	D	18	D	28	B	38	D	48	C	58	B
9	A	19	C	29	A	39	C	49	D	59	A
10	B	20	B	30	B	40	B	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) OC and OD are opposite rays, so $\angle AOC + \angle AOD = 180$: $(2x+18)+(3x-8)=180 \rightarrow 5x+10=180 \rightarrow x=34$. $\angle AOD = 3(34)-8 = 94^\circ$. $\angle BOC$ is vertically opposite $\angle AOD$, so $\angle BOC = 94^\circ$. The value 86° is the vertical of $\angle AOC$ ($2 \cdot 34+18$) instead; 34° is x ; 46° wrongly computes $180-134$.

2. (C) Vertically opposite angles are equal: $5x-4 = 3x+30 \rightarrow 2x=34 \rightarrow x=17$. The angle is $5(17)-4 = 81^\circ$. Its linear-pair neighbour = $180 - 81 = 99^\circ$. The value 81° stops at the vertical angle; 17° is x ; 162° doubles 81 instead of subtracting from 180.

3. (C) $\angle BOD = \angle AOC = 4(12)+9 = 57^\circ$. OE splits $\angle BOD$ into $\angle BOE = 31^\circ$ and $\angle DOE$, so $\angle DOE = 57 - 31 = 26^\circ$. Taking 57° forgets to subtract $\angle BOE$; 31° re-reads the given; 88° adds $57+31$.

4. (A) Adjacent angles form a linear pair: $3x + (x+40) = 180 \rightarrow 4x = 140 \rightarrow x = 35$. The two values are $3x = 105^\circ$ and $x+40 = 75^\circ$. The larger is 105° . The value 75° is the smaller; 35° is x ; 115° comes from $x+80$ in error.

5. (B) $(6x-10)+(4x+30)=180 \rightarrow 10x+20=180 \rightarrow x=16$. The angles are $6(16)-10 = 86^\circ$ and $4(16)+30 = 94^\circ$. Difference = $94 - 86 = 8^\circ$. The value 16° is x ; 94° is only the larger angle; 180° is the sum.

6. (D) Vertical angles are equal: $7x-16 = 5x+22 \rightarrow 2x=38 \rightarrow x=19$. $\angle 1 = 7(19)-16 = 117^\circ$. The other vertical pair is the linear-pair neighbour: $180 - 117 = 63^\circ$. The value 117° is $\angle 1$ itself; 19° is x ; 126° doubles 63 instead of using it directly.

7. (C) Angles at a point sum to 360: $(2x+5)+(3x-15)+(4x+10)=360 \rightarrow 9x+0=360 \rightarrow x=40$. The angles are 85° , 105° , 170° . The largest is $4(40)+10 = 170^\circ$. The values 85° and 105° are smaller angles; 150° drops the +10.

8. (D) Sum at a point = 360: $2y + 3y + (y+60) = 360 \rightarrow 6y + 60 = 360 \rightarrow 6y = 300 \rightarrow y = 50$. Angles: $2y = 100^\circ$, $3y = 150^\circ$, $y+60 = 110^\circ$. Smallest = 100° . Its supplement = $180 - 100 = 80^\circ$. The value 100° is the angle itself; 150° is the largest; 30° mis-subtracts.

9. (A) Sum at a point = 360: $a + (a+10) + (a+20) + (a+30) = 360 \rightarrow 4a + 60 = 360 \rightarrow 4a = 300 \rightarrow a = 75$. The largest is $a+30 = 105^\circ$. The value 75° is the smallest (a); 95° is $a+20$; 90° is the mean, not any actual angle.

10. (B) Five equal angles around a point: each = $360 \div 5 = 72^\circ$. The bisector splits one into two equal parts of $72 \div 2 = 36^\circ$ each. The value 72° forgets to bisect; 45° wrongly halves a right angle; 18° bisects twice.

11. (C) Sum at a point = 360: $3x + 90 + 2x + (x+30) = 360 \rightarrow 6x + 120 = 360 \rightarrow 6x = 240 \rightarrow x = 40$. $\angle AOB = 3(40) = 120^\circ$. The value 40° is x ; 80° is $\angle COD$ ($2x$); 70° is $\angle DOA$ ($x+30$).

12. (C) Sum at a point = 360: $(3x-10)+(x+40)+(2x+30)=360 \rightarrow 6x+60=360 \rightarrow 6x=300 \rightarrow x=50$. Angles: $3x-10 = 140^\circ$, $x+40 = 90^\circ$, $2x+30 = 130^\circ$. The middle-sized is 130° . The value 140° is the largest; 90° is the smallest; 50° is x .

13. (B) Corresponding angles between parallel lines are equal: $5x-12 = 3x+8 \rightarrow 2x=20 \rightarrow x=10$. The angle = $5(10)-12 = 38^\circ$. Its linear-pair neighbour = $180 - 38 = 142^\circ$. The value 38° is the angle itself; 10° is x ; 122° mis-subtracts ($180-58$).

14. (C) Corresponding angles equal: $4x+15 = 6x-25 \rightarrow 40 = 2x \rightarrow x=20$. The angle = $4(20)+15 = 95^\circ$. Its co-interior partner is supplementary: $180 - 95 = 85^\circ$. The value 95° equates co-interior instead of supplementing; 20° is x ; 105° mis-subtracts.

15. (C) The corresponding angle at Q equals 118° (lines parallel). Its vertically opposite angle is also 118° . So the answer is 118° . The value 62° takes the linear pair instead of the vertical; 236° doubles; 90° assumes a perpendicular crossing.

16. (B) Corresponding angles equal: $2x+30 = 4x-10 \rightarrow 40 = 2x \rightarrow x=20$. The angle = $2(20)+30 = 70^\circ$. Its linear-pair neighbour = $180 - 70 = 110^\circ$. The value 70° stops at the angle; 20° is x ; 140° doubles 70.

17. (A) The 64° interior angle and the same-side interior angle at m are co-interior, hence supplementary: $180 - 64 = 116^\circ$. The value 64° wrongly equates co-interior angles; 26° subtracts from 90; 128° doubles 64.

18. (D) Corresponding angles equal: $x+25 = 2x-15 \rightarrow x=40$. The angle = $40+25 = 65^\circ$. Its vertically opposite angle is equal, 65° . The value 115° takes the linear pair; 40° is x ; 50° mis-computes.

19. (C) Alternate interior angles are equal: $3x-5 = x+41 \rightarrow 2x=46 \rightarrow x=23$. The angle = $3(23)-5 = 64^\circ$. Its co-interior partner (linear pair at the crossing) = $180 - 64 = 116^\circ$. The value 64° is the alternate angle; 23° is x ; 46° mis-computes.

20. (B) Alternate exterior angles are equal: $5x-30 = 3x+10 \rightarrow 2x=40 \rightarrow x=20$. The angle = $5(20)-30 = 70^\circ$. Its linear-pair neighbour = $180 - 70 = 110^\circ$. The value 70° is the angle itself; 20° is x ; 40° mis-computes.

21. (A) Alternate interior angles are equal, so $\angle A = \angle B$. Substituting: $\angle A = 2\angle A - 24 \rightarrow \angle A = 24^\circ$. The value 48° doubles; 12° halves; 156° takes the supplement.

22. (A) Alternate interior angles equal: $7x-4 = 5x+14 \rightarrow 2x=18 \rightarrow x=9$. The angle = $7(9)-4 = 59^\circ$. Its co-interior partner = $180 - 59 = 121^\circ$. The value 59° is the alternate angle; 9° is x ; 31° subtracts from 90.

23. (B) The exterior 142° angle has an interior linear-pair neighbour of $180 - 142 = 38^\circ$ at the same crossing; that interior angle's alternate interior angle at the lower line equals it, 38° . The value 142° forgets the linear-pair step; 52° subtracts from 90 wrongly; 48° mis-computes.

24. (D) Alternate exterior angles equal: $3x+15 = 4x-5 \rightarrow x=20$. The angle = $3(20)+15 = 75^\circ$. Its linear-pair neighbour = $180 - 75 = 105^\circ$. The value 75° is the exterior angle; 20° is x ; 95° mis-computes.

25. (B) Co-interior angles are supplementary: $(3x+20)+(2x+10)=180 \rightarrow 5x+30=180 \rightarrow x=30$. The angles are $3(30)+20 = 110^\circ$ and $2(30)+10 = 70^\circ$; larger = 110° . Its alternate interior angle equals it, 110° . The value 70° is the smaller co-interior angle; 30° is x ; 100° mis-computes.

26. (D) Co-interior angles are supplementary: $7k + 2k = 180 \rightarrow 9k = 180 \rightarrow k = 20$. Angles: $7k = 140^\circ$, $2k = 40^\circ$. Smaller = 40° . The value 140° is the larger; 20° is k ; 90° wrongly assumes equality.

27. (A) Let the smaller be t ; the larger is $4t$. Co-interior angles are supplementary: $4t + t = 180 \rightarrow 5t = 180 \rightarrow t = 36$. Larger = $4(36) = 144^\circ$. The value 36° is the smaller; 45° and 135° come from wrongly treating them as equal halves or a 3:1 split.

28. (B) Co-interior angles sum to 180: $a + b = 180$ and $a - b = 36$. Adding: $2a = 216 \rightarrow a = 108^\circ$. Larger = 108° . The value 72° is the smaller; 36° is the difference; 90° assumes equality.

29. (A) Co-interior angles are supplementary: $5x + (4x-9) = 180 \rightarrow 9x - 9 = 180 \rightarrow 9x = 189 \rightarrow x = 21$. The $(5x)^\circ$ angle = 105° . Its corresponding angle at the other line equals it, 105° . The value 75° is the other co-interior angle $(4x-9)$; 21° is x ; 115° mis-computes.

30. (B) The two angles form a linear pair, so they sum to 180 in ratio 3:2: $3k + 2k = 180 \rightarrow 5k = 180 \rightarrow k = 36$. The alternate-interior (3 parts) = 108° . The value 72° is the 2-part angle; 54° halves wrongly; 120° uses a 2:1 split.

31. (D) Corresponding angles equal: $8x-12 = 5x+21 \rightarrow 3x=33 \rightarrow x=11$. $\angle Q = 5(11)+21 = 76^\circ$. Its linear-pair neighbour = $180 - 76 = 104^\circ$. The value 76° is $\angle Q$ itself; 11° is x ; 94° mis-computes.

32. (B) Linear pair: $7x + 2x = 180 \rightarrow 9x = 180 \rightarrow x = 20$. $\angle AOC = 7(20) = 140^\circ$. The bisector halves it: $\angle DOC = 140 \div 2 = 70^\circ$. The value 140° forgets to bisect; 40° is $\angle COB$ ($2x$); 20° is x .

33. (D) Alternate interior angles equal: $2x+9 = 3x-6 \rightarrow x=15$. The angle = $2(15)+9 = 39^\circ$. Its co-interior partner = $180 - 39 = 141^\circ$. The value 39° is the alternate angle; 15° is x ; 151° mis-subtracts.

34. (D) Co-interior angles supplementary: $(x+30) + 2x = 180 \rightarrow 3x = 150 \rightarrow x = 50$. Angles: $x+30 = 80^\circ$, $2x = 100^\circ$; larger = 100° . Its vertically opposite angle equals it, 100° . The value 80° is the smaller; 50° is x ; 130° mis-computes.

35. (C) Vertical angles equal: $4x+5 = 6x-25 \rightarrow 30 = 2x \rightarrow x=15$. $\angle AOC = 4(15)+5 = 65^\circ$. $\angle AOD = 180 - 65 = 115^\circ$. The value 65° is $\angle AOC$; 15° is x ; 130° doubles 65.

36. (A) Corresponding angles equal: $6x-8 = 4x+22 \rightarrow 2x=30 \rightarrow x=15$. The angle = $6(15)-8 = 82^\circ$. Its co-interior partner = $180 - 82 = 98^\circ$. The value 82° is the angle itself; 15° is x ; 108° mis-computes.

37. (C) Corresponding angles between parallel lines must be EQUAL. Here $88^\circ \neq 92^\circ$, so the lines are not parallel. Treating 88° and 92° as 'close enough' or as supplementary ($88+92=180$) wrongly suggests parallel; neither line is perpendicular nor coincident.

38. (D) For parallel lines, co-interior angles must sum to exactly 180° . Here $95 + 86 = 181^\circ \neq 180^\circ$, so the lines are not parallel. Reading the test as 'equal' or 'sums to 180' is the

trap; nearness to 90° is irrelevant.

39. (C) Alternate interior angles between parallel lines must be EQUAL. Since $70^\circ \neq 76^\circ$, the lines are not parallel. The 6° difference does not give the angle of meeting; nothing here makes them perpendicular.

40. (B) Co-interior angles summing to exactly 180° ($105 + 75 = 180$) is the test for parallel lines, so the lines ARE parallel. Mistaking the test for 'equal angles' would wrongly reject parallelism; the crossing is not perpendicular.

41. (A) Equal corresponding angles guarantee parallel lines, so the lines are parallel. Equal corresponding angles never indicate the crossing angle of intersecting lines; perpendicular would need 90° .

42. (B) $\angle 2 = \angle 1 = 124^\circ$ (corresponding). $\angle 3 = 180 - 124 = 56^\circ$ (linear pair). $\angle 4 = \angle 3 = 56^\circ$ (vertically opposite). The value 124° stops at $\angle 2$; 34° subtracts from 90 ; 236° doubles incorrectly.

43. (B) Because the three lines are parallel, the same-side interior angle is co-interior with the 73° angle and is supplementary: $180 - 73 = 107^\circ$. (Going through the middle line m changes nothing — corresponding then co-interior returns the same supplement.) The value 73° wrongly equates co-interior; 17° subtracts from 90 ; 146° doubles 73 .

44. (A) Draw a line through E parallel to AB and CD . It splits $\angle BEC$ into two parts: the part adjacent to BE equals $\angle ABE = 40^\circ$ (alternate interior with AB), and the part adjacent to CE equals $\angle DCE = 65^\circ$ (alternate interior with CD). So $\angle BEC = 40 + 65 = 105^\circ$. The value 25° subtracts; 75° uses $40+35$; 140° doubles 70 .

45. (D) When both given angles fall on the same side of the helper line through E (parallel to AB and CD), $\angle BEC$ is their difference: $58 - 23 = 35^\circ$. The value 81° adds them; 116° doubles 58 ; 46° doubles 23 .

46. (C) The corresponding angle at Q (lower-right) is 118° . Its co-interior partner — the lower-left interior angle at Q — is supplementary: $180 - 118 = 62^\circ$. The value 118° wrongly equates co-interior angles; 236° doubles; 28° subtracts from 146 .

47. (D) The helper line through E gives $\angle BEC = \angle ABE + \angle DCE$. So $90 = 35 + \angle DCE \rightarrow \angle DCE = 55^\circ$. The value 125° adds instead of subtracting ($35+90$); 45° complements 45 ; 35° re-reads $\angle ABE$.

48. (C) Co-interior angles sum to 180° , so their halves sum to 90° . In the triangle formed by the transversal segment and the two bisectors, the third angle = $180 - 90 = 90^\circ$. The bisectors always meet at 90° . The value 45° halves once too often; 100° and 80° fix on a particular co-interior value.

49. (D) The bisector splits the 84° angle (between the transversal and line b) into two equal 42° parts, so it makes 42° with line b. The value 84° forgets to bisect; 48° bisects the supplement; 96° doubles 48.

50. (C) $\angle Q$ is co-interior with 73° , so $\angle Q = 180 - 73 = 107^\circ$. Its bisector makes $107 \div 2 = 53.5^\circ$ with line m. The value 36.5° bisects 73 instead of 107; 73° forgets the supplement and the bisection; 107° forgets to bisect.

51. (D) Linear pair: $7x + 2x = 180 \rightarrow x = 20$, so $\angle AOC = 140^\circ$ and $\angle COB = 40^\circ$. OD bisects $\angle AOC \rightarrow \angle DOC = 70^\circ$. $\angle DOB = \angle DOC + \angle COB = 70 + 40 = 110^\circ$. The value 70° is only $\angle DOC$; 40° is $\angle COB$; 140° is the whole $\angle AOC$.

52. (D) Each bisector makes 48° with the transversal, and because the alternate interior angles are equal (96°) and lie on opposite sides, their bisectors are themselves a pair of equal alternate interior angles (48°) — so the bisectors are parallel. The 96° and 48° options confuse the original or bisected angle with the angle between the bisectors; they are not perpendicular.

53. (D) At a crossing the angles in cyclic order must alternate a, b, a, b with $a + b = 180$ and $a \neq b$ (non-perpendicular). 65, 115, 65, 115 fits. The orderings 65,115,115,65 and 65,65,115,115 place equal (vertical) angles adjacent, which is impossible; all 90s is the perpendicular case.

54. (A) Co-interior (same-side interior) angles between parallel lines are SUPPLEMENTARY (sum 180°), not equal — so that statement is false. The other three are all true standard facts.

55. (A) Equal co-interior angles only force parallelism if they also sum to 180° , i.e. only when each is 90° ; in general 'equal co-interior angles' does NOT guarantee parallel lines. The other three are valid parallel tests.

56. (C) If one angle is 90° , its linear-pair neighbour is also 90° , making all four angles at that crossing right angles; corresponding angles at the parallel line copy them, so all eight are right angles. The values 4, 2, 1 stop the chain too early.

57. (A) A perpendicular crossing makes every one of the eight angles 90° , so there is exactly 1 distinct measure. The values 2 and 4 are the distinct-measure counts for a slanted transversal; 8 assumes all differ.

58. (B) Since $c \parallel b$ and $a \perp b$, line a is also perpendicular to c. A transversal making 35° with a therefore makes $90 - 35 = 55^\circ$ with c. The value 35° forgets the perpendicular turn; 90° is the a–c angle; 125° is the obtuse (non-acute) supplement.

59. (A) Because $a \parallel b$ and the transversal is perpendicular to a, it is also perpendicular to b, so $(5x - 5)^\circ = 90^\circ \rightarrow 5x = 95 \rightarrow x = 19$. The values 18 and 17 come from solving $5x - 5 =$

85 or 80; 95 confuses x with the angle.

60. (B) Two lines each perpendicular to b are parallel to each other, so $a \parallel c$. A transversal therefore makes the SAME angle, 28° , with c as with a . The value 62° is the complement; 90° treats a and c as perpendicular; 0° confuses 'parallel lines' with 'zero angle to the transversal'.